

Module 04: Cognition in High School

Learning Objectives

1. Define **Cognition** within the context of High School.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In High School, we see Cognition manifest in: * **Specific Example 1:** A large language model like ChatGPT performs cognition as Active Inference: given a prompt, it generates the next token by predicting which word minimizes surprise given all previous context -- its billions of parameters encode a massive generative model of language, and the "temperature" setting literally controls how much prediction error (randomness) the model tolerates in its cognitive output. * **Specific Example 2:** Spam filters in your email demonstrate automated cognition through Active Inference: the filter maintains a generative model of what legitimate email looks like (learned from millions of labeled examples), compares each incoming message against that model, and flags messages with high prediction error (unusual sender patterns, suspicious links) as spam -- continuously updating its beliefs as new spam tactics emerge.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.